

# HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

## PROFESSIONAL EXPERIENCE

**SOFTWARE ENGINEER – HCL Technologies | Chennai, India**

**09/2021 – 01/2025 (3.5 Years)**

- Developed and deployed production-grade **Python** agentic systems for critical workflows, managing over **10,000 daily transactions**.
- Implemented advanced **context management** strategies across multi-step agent executions, ensuring state consistency and reducing error rates by **15%**.
- Built and maintained **RAG pipelines** that ingest and process structured, semi-structured, and unstructured data from diverse sources, enhancing data retrieval accuracy by **20%**.
- Designed and implemented agent workflows using **LangGraph**, orchestrating complex task sequences with robust **state management** and **idempotency** controls.
- Integrated **relational databases** (PostgreSQL) and **vector stores** to support agentic system data persistence and retrieval, improving query performance by **30%**.
- Automated ETL processes and scheduled reporting using **Python** scripts, reducing manual data preparation time by **40 hours** per month.

## SKILLS

**AI/ML** – AI, Agentic Systems, RAG, Embeddings, Retrieval Strategies, LangGraph

**Programming Languages** – Python, SQL, Java, R

**Databases** – PostgreSQL, MySQL, Oracle, NoSQL Databases, Vector Stores

**Frameworks/Libraries** – LangGraph, Spring Boot, pandas, NumPy, scikit-learn

**Cloud/DevOps** – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

**Data Visualization/BI** – Power BI, DAX, Tableau, Power Query

**Methodologies** – Agile, Scrum, Change Management

## EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of  
Ireland, Dublin**

**02/2026**

**BACHELOR OF ENGINEERING in Computer Science – SNS College  
of Technology, Coimbatore, India**

**01/2021**

## PROJECTS

**Agentic Workflow Orchestrator for Financial Operations**

- Developed a Python-based agentic system to automate transaction reconciliation workflows, processing over **5,000 reconciliation tasks** daily.
- Implemented stateful orchestration using **LangGraph** for multi-step analysis of payment messages, ensuring **99.95% accuracy** in identifying discrepancies.

- Integrated **PostgreSQL** for storing transaction states and results, and utilized **FastAPI** to expose operational dashboards for monitoring.

#### **Intelligent Document Processing Pipeline**

- Constructed a **RAG pipeline** to ingest and analyze enterprise documents for regulatory compliance checks, reducing manual review time by **50%**.
- Applied **Python** with libraries like `pandas` and `scikit-learn` for data preprocessing and feature extraction from semi-structured reports.
- Designed embedding strategies and retrieval mechanisms to efficiently query a knowledge base of over **10,000 compliance documents**.

### **ACHIEVEMENTS & CERTIFICATIONS**

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)